

THE
ART
OF
COLLECTION



標本與收藏學

The Art of Collection

Preserving nature's secrets for future generations
為後代保存大自然的秘密

ICS Creator Academy · Summer 2026



What is a Specimen? 什麼是標本？

A specimen is a sample of a plant, animal, or mineral preserved for scientific study 用於科學研究的保存樣本



Plants 植物標本

Pressed flat, dried, and mounted on paper. Some specimens are 400 years old but still show the plant's original colour! 有些標本400年了但還看得到顏色！



Insects 昆蟲標本

Pinned and labelled on boards. The Natural History Museum London has 80 million insect specimens — the largest collection on Earth! 倫敦自然史博物館有8000萬件！



Skeletal 骨骼標本

Bones carefully cleaned and assembled. Scientists can tell age, diet, disease, and cause of death from bones alone! 只靠骨頭就能知道年齡、飲食和死因！



Wet specimens 濕標本

Animals preserved in alcohol or formaldehyde in glass jars. Darwin's specimens from the Beagle voyage are kept this way! 達爾文的標本就是這樣保存的！



Minerals 礦石標本

Rocks, crystals, and gems. The Smithsonian's Hope Diamond 希望之鑽 is the world's most famous mineral specimen — 45.52 carats! 全球最著名的礦石標本！



Shell & fossil 貝殼與化石

Shells, fossils, and coral tell us about ancient oceans. Some fossil specimens are 500 million years old! 有些化石標本有5億年歷史！



Why Do We Collect Specimens? 為什麼要製作標本？



Scientific study 科學研究

Specimens allow scientists to study organisms in detail long after they're gone. Many species were only identified from museum specimens! 許多物種只能從博物館標本研究！



Historical record 歷史記錄

Specimens are time capsules! Darwin's beetles collected in 1835 are still being studied today. They contain DNA we can analyse! 達爾文1835年的甲蟲至今仍在被研究！



Track changes 追蹤變化

By comparing old and new specimens, scientists can track climate change, pollution, and evolution over centuries! 比較新舊標本可以追蹤氣候變化！



Extinct species 已滅絕物種

Specimens of extinct animals let us study creatures we can never see alive. The Dodo 渡渡鳥 only exists in museums now! 已滅絕的動物只能在博物館看到！



Medicine 醫學

Many medicines come from plants and animals discovered through specimen collections! Penicillin 青黴素 was found from a mould specimen! 青黴素從黴菌標本發現的！



Conservation 保育

Specimen DNA can help revive endangered species! Scientists are working on bringing back the Woolly Mammoth 長毛象 using ancient DNA! 古代DNA可能讓長毛象復活！



The beetle obsession 甲蟲迷

Darwin loved beetles so much he once put a LIVE beetle IN HIS MOUTH when his hands were full — just so he wouldn't lose it! 他把活甲蟲放進嘴巴！



The famous finches 達爾文雀

He collected 13 finches from the Galápagos Islands. Their different beak shapes inspired his theory of natural selection 天擇說 — changing science forever!



Plant specimens 植物標本

Over 1,500 plants carefully pressed and labelled. Many are still in the Natural History Museum London today — 190 years later! 190年後仍在博物館保存！



Giant fossil bones 巨大化石

In Argentina, he found fossils of giant sloths and armadillos. Proof that giant creatures once roamed South America! 南美洲曾有巨型生物！



查爾斯·達爾文

Charles Darwin

1809–1882
Naturalist & Collector



5-year voyage on HMS Beagle 1831-1836
Collected over 5,000 specimens!
5年航行收集超過5000件標本！



World's Greatest Specimen Collections 世界最偉大的標本收藏

Natural History Museum

倫敦自然史博物館

London, UK

GB

80 million specimens

8千萬件標本

 Their famous blue whale skeleton is 25 metres long! Discovered 1891, displayed since 2017. 25公尺長的藍鯨骨骼！

Smithsonian Natural History

美國史密森尼自然史博物館

Washington D.C., USA

US

148 million specimens

1.48億件標本

 Home of the Hope Diamond 希望之鑽 — 45.52 carats, worth \$250 million! 45.52克拉 · 價值2.5億美元！

Muséum National

法國自然歷史博物館

Paris, France

FR

67 million specimens

6700萬件標本

 Has a 'Gallery of Evolution' with hundreds of taxidermied animals in dramatic poses! 戲劇性姿勢的動物標本走廊！

National Museum of Natural History

中國國家自然博物館

Beijing, China

CN

370,000+ specimens

37萬件標本

 Has the most complete dinosaur fossil collection in Asia! 亞洲最完整的恐龍化石收藏！



Surprising Specimen Stories 驚人的標本故事



Jumbo the Elephant 大象鄧波

The world's most famous elephant, died 1885. P.T. Barnum had him taxidermied 製成標本 and displayed. Jumbo is now at Tufts University in the USA!



The oldest seed 最老的種子

A 2,000-year-old Judean date palm seed found in Israel was successfully GERMINATED in 2005! Dead for 2,000 years — then planted and grew! 沉睡2000年的種子成功發芽！



The last Dodo 最後一隻渡渡鳥

The Dodo went extinct in 1681. Only a few specimen parts remain worldwide. Scientists are trying to extract DNA to bring it back! 科學家正嘗試用DNA讓牠復活！



The Coelacanth mystery 腔棘魚謎題

Scientists thought Coelacanth 腔棘魚 was extinct for 65 million years — until a live one was caught in 1938! Museum specimens helped identify it! 博物館標本幫助確認了這條魚！

If you could preserve ONE thing for future scientists 1,000 years from now, what would you choose? 你想為1000年後的科學家保存什麼？



Let's Discuss! 來討論！

1

If you found a new insect species, what would you name it? 如果你發現了一個新昆蟲物種，你會給它取什麼名字？

2

Should scientists try to bring back extinct animals like the Dodo or Mammoth? Why? 應該讓已滅絕的動物復活嗎？

3

What animal or plant would you most like to study as a specimen? 你最想研究哪種動物或植物的標本？

4

Why is it important to preserve specimens for future generations? 為什麼要為未來的人保存標本？